Networking Assignment

Computer Networking & IP Addressing Solutions

Question 1: Differences Between IPv4 and IPv6

Task: List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Answer:

IPv4 Address	IPv6 Address
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in decimal format.	Written in hexadecimal format.
Example: 192.168.1.10	Example: 2001:db8::10
Uses four octets separated by dots.	Uses eight groups separated by colons (groups can be shortened).
Provides about 4.3 billion addresses.	Provides a vastly larger address space.
Supports broadcast.	Does not use broadcast; it uses multicast and anycast.

Conclusion: IPv6 provides a much larger address space than IPv4 and was designed to support the growing number of internet-connected devices.

Question 2: Assign Static IPv4 Addresses to Four Devices

Task: Assign a unique static IPv4 address to a laptop, phone, smart TV, and printer using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Answer:

- **Network:** 192.168.10.0/24
- **Subnet mask:** 255.255.255.0
- **Default gateway:** 192.168.10.1

Device	Static IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Conclusion: Each device has a unique IP address in the same subnet and uses the router's IP address as its default gateway.

Question 3: Calculate Subnets and Hosts Using /27

Task: Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges and usable host IPs in each.

Answer:

Given:

• Network: 192.168.1.0/24
• New Subnet Mask: /27 (255.255.255.224)

Calculations:

• Bits borrowed: 27 - 24 = 3 bits
• Number of subnets: 2^3 = 8 subnets
• Host bits remaining: 32 - 27 = 5 bits
• Addresses per subnet: 2^5 = 32 addresses
• Usable hosts per subnet: 32 - 2 = 30 hosts

Metric	Value
Subnets in the /24 network	8
Usable hosts per subnet	30

First Three Subnet Ranges:

Subnet	Network Address	Usable Host IP Range	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Conclusion: A /27 subnet provides 8 equal-sized subnets within a /24 network, with 30 usable host addresses in each.

Question 4: Find Network Address, Broadcast Address, and Valid Hosts

Task: Given IP address 10.0.5.17 with subnet mask 255.255.255.0, determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Answer:

Given:

• IP Address: 10.0.5.17
• Subnet Mask: 255.255.255.0
• CIDR Notation: /24

Item	Answer
Network address	10.0.5.0
Broadcast address	10.0.5.255
First usable host	10.0.5.1
Last usable host	10.0.5.254
Total usable hosts	254

Explanation: The /24 mask means the first 24 bits identify the network and the remaining 8 bits identify hosts. Performing bitwise AND operation gives the network address (10.0.5.0). Setting all host bits to 1 gives the broadcast address (10.0.5.255).

Conclusion: The valid host range is 10.0.5.1 to 10.0.5.254.

Question 5: Verify the Calculation Using an Online Subnet Calculator

Task: Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Answer:

Verification Steps:

1. Open the Subnet Calculator (e.g., https://www.subnet-calculator.com/).
2. Enter IP address 10.0.5.17.
3. Select subnet mask 255.255.255.0 or CIDR /24.
4. Calculate the results.
5. Take a screenshot showing the entered IP and calculated network details.

One-line observation:

“The subnet calculator confirmed my manual answer: network address 10.0.5.0, broadcast address 10.0.5.255, and usable host range 10.0.5.1–10.0.5.254.”

Note for Submission: Please take and attach your own screenshot of the calculator page directly below this section before final submission.